Intelligent Sports and Health 1 (2025) 210–222

Contents lists available at ScienceDirect

Intelligent Sports and Health

journal homepage: www.keaipublishing.com/en/journals/intelligent-sports-and-health

Review article

Multimodal physiological signals from wearable sensors for affective computing: A systematic review☆

Fang Li[1], Dan Zhang[2,*]

Department of Psychological and Cognitive Sciences, Tsinghua University, Beijing 100084, China

ARTICLE INFO

Keywords:
Affective computing
Wearable sensors
Multimodal
Deep learning

ABSTRACT

In recent years, advances in wearable devices have driven the rapid development of multimodal physiological signals for affective computing. In this paper, we systematically review 34 relevant studies between 2015 and 2025 following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines, focusing on the combination of multimodal wearable sensing with learning models. We find that multimodal measurements have become increasingly common in the field, with electrodermal activity (EDA) and electrocardiograms (ECGs) being the most commonly used signal modalities. Several widely used public datasets, including WESAD, DAPPER, and DEAP, provide a basis for model optimization and comparative experiments. Deep learning methods have been widely adopted, with convolutional neural networks (CNNs) and long short-term memory (LSTM) being the most commonly used, and gradually extending to architectures such as the transformer. Despite recent advancements in multimodal emotion recognition studies, the field still faces key challenges, including the lack of long-term continuous data in naturalistic settings, the limited adaptability of current models to multimodalities, and the limited generalizability and interpretability of models for real-world applications. Future research needs to continue to address data acquisition, method optimization, and model generalizability in applications.

1. Introduction

Affective computing, which was proposed by Picard in 1997 [1], aims to endow computational systems with the ability to recognize, interpret, and respond to human emotions. Early studies relied primarily on behavioural cues such as facial expressions and voice tone for modelling affective states [2–5]. In recent years, physiological signals have emerged as powerful alternatives because of their noninvasiveness, continuity, objectivity, and nonlanguage dependence characteristics. Physiological signals, such as electroencephalograms (EEGs) and electrodermal activity (EDA), have been used in various scenarios, including mental health monitoring, intelligent tutoring systems, athletic training, and emotion-aware human–computer interactions [6–9,10].

With the rapid advancement of wearable sensing technologies, physiological affective computing has entered a new phase. Wearable devices are capable of continuously acquiring multimodal physiological signals from multiple sensor channels that differ in terms of sampling frequency, physiological origin, and signal characteristics. These capabilities create significant application opportunities. For instance, in sports, multimodal physiological signals can be leveraged to monitor athletes' real-time physiological responses, supporting emotional regulation by effectively managing stress and anxiety during competitions [11]. Furthermore, detecting fatigue-related emotional shifts through multimodal physiological signals enables targeted optimization of athletes' performance [12]. Multimodal sensors offer new opportunities for richer emotional inference but also present significant challenges in signal synchronization, noise reduction, data fusion, and model construction. Against this backdrop, a systematic review focused on multimodal physiological signals from wearable devices, especially from a computational perspective, is both timely and necessary.

In this study, we focus on studies of affective computing based on multimodal physiological signals collected by wearable devices

☆ This work is supported by a grant from the General Office of the National Language Commission Research Planning Committee (ZDA145–18), the Graduate Education Innovation Grants of Tsinghua University (202504Z005), the Education Innovation Grants of Tsinghua University (DX02_20), and the Capital's Funds for Health Improvement and Research (CFH2024–4–21210).

* Corresponding author.
E-mail addresses: li-f24@mails.tsinghua.edu.cn (F. Li), dzhang@tsinghua.edu.cn (D. Zhang).
[1] 0000–0002-7423–0090
[2] 0000–0002-7592–3200

https://doi.org/10.1016/j.ish.2025.10.001
Received 28 July 2025; Received in revised form 17 September 2025; Accepted 9 October 2025
Available online 29 October 2025

published during the past ten years. We systematically review the data processing flow, multimodal fusion strategy, and model architecture. We also summarize the development status and challenges of deep learning methods in the field of affective computing.

Several recent reviews in this area have been published. Each review concentrated on specific application domains of affective computing. For instance, Saganowski et al. [13] conducted a comprehensive review on the use of physiological signals for emotion recognition, mainly discussing the differences between laboratory and real-world scenarios, as well as feature selection methods for emotion recognition. Machhi and Sha [14] presented an extensive review of wearable devices applied in affective computing, focusing primarily on device types, sensor technologies, and associated physiological parameters. Shen and Hu [15] conducted a systematic review of wearable-based emotion sensing within educational contexts, highlighting the types of physiological signals and learning scenarios involved. Hickey et al. [16] conducted a systematic review of smart devices and wearable technologies for detecting and monitoring mental health conditions and stress, offering broad insight into device types, monitored parameters, and clinical applications. Several studies have focused on signal characteristics and feature extraction methods using single-modal physiological signals [17–19]. While several recent reviews have addressed the individual aspects of wearable emotion recognition, few have systematically examined the full computational pipeline, spanning data acquisition and preprocessing, fusion strategy design, and deep learning model architecture with performance analysis, specifically for multimodal physiological signals in wearable-based settings.

This review systematizes and summarizes recent studies on affective computing using multimodal physiological signals collected via wearable devices. It is intended to serve as a comprehensive reference for researchers studying emotion recognition in everyday settings, especially those interested in combining multimodal physiological data with novel computational modelling approaches. The remainder of this paper is organized as follows. Section 2 describes the methodology employed. Section 3 presents the results, covering major stages of the research pipeline, including data acquisition, signal modalities, data processing and feature representations, multimodal fusion strategies, modelling architectures, and validation procedures. Section 4 discusses future research directions, and Section 5 presents the conclusions drawn from the analysis.

The contributions of this paper are as follows.

1. **Systematic Literature Review (SLR)**. This review systematically analyses 34 empirical studies that employed wearable multimodal physiological signals for emotion recognition, quantifying the computational pipeline from data acquisition to model validation.
2. **Signal Types and Dataset Statistics**. Statistical information on six core dimensions of commonly used datasets (such as WESAD, DAPPER, etc.) and modalities (such as EDA, ECG, and SKT, etc.) is summarized and quantified, offering a valuable reference for researchers to select suitable datasets and understand their applicability.
3. **Fusion Model Validation Pipeline**. From a signal-driven perspective, we analyse methods for processing physiological signals with complex noise and labelling delays, systematically examine multimodal fusion strategies, and clarify the development trend of models from machine learning to typical deep learning (such as CNN, LSTM, etc.), continuing on to novel architectures such as transformers. This analysis provides the field with an overall understanding of how signal characteristics, fusion strategies, and modelling methods jointly affect emotion recognition performance.

2. Methodology

We followed procedures in the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) statement [20] to systematically investigate how multimodal signals from wearable systems have been applied to affective computing. A detailed PRISMA flow diagram is shown in Fig. 1. This review focuses on identifying empirical studies that integrate multiple physiological signals using computational models for emotion-related tasks. We queried four major academic databases, PubMed, IEEE Xplore, Scopus, and Web of Science, covering literature published between January 2015 and May 2025. We selected this period because of rapid developments in wearable sensing and deep learning during this time [21]. Notably, 30 of the 34 studies reviewed were published after 2020, reflecting the growing research interest and methodological innovation over the past 10 years.

To retrieve relevant articles, we developed a Boolean search strategy centred on three core dimensions: (1) affective conceptualization (e.g., emotion, mood, and sentiment), (2) multimodal sensing or signal fusion, and (3) wearable sensors. We ran the following query: (affective OR emotion OR mood) AND (multimodal OR fusion OR ("multiple signals")) AND (wearable OR ("personal device") OR ("smart device")). Database-specific syntax and filters were adapted where necessary (e.g., MeSH terms for PubMed, indexing terms for IEEE). Duplicates were removed, and all remaining articles were evaluated using predefined inclusion and exclusion criteria as described in the next section.

To ensure the relevance and quality of the included studies, we applied a set of inclusion and exclusion criteria during the screening process. Studies were considered eligible if they met all of the following conditions:

(1) Publication year between January 2015 and May 2025 and written in English.
(2) Empirical research involving wearable devices that collected at least two types of physiological signals (e.g., HR and EDA).
(3) Empirical research focusing on emotion recognition, affective state monitoring, or related tasks.
(4) Empirical research using computational models to process physiological data.
(5) Empirical research reporting quantitative evaluation metrics such as accuracy, F1 score, sensitivity, or specificity.

The exclusion criteria were defined as follows:

(1) Research using single-modality signals (e.g., electrocardiogram (ECG) only or EDA only).
(2) Data acquisition via nonwearable devices (e.g., fMRI).
(3) Research using behavioural, environmental, or audio-visual modalities (e.g., facial expressions, voice, posture, and movement).
(4) Research focusing on brain imaging modalities such as EEG, functional near-infrared spectroscopy (fNIRS), and functional magnetic resonance imaging (fMRI).
(5) Non-peer-reviewed studies, review studies, conference abstracts without full texts, or those written in languages other than English.

After removing duplicates and applying inclusion/exclusion criteria, 34 studies were retained for full analysis. Author disciplines included computer science and engineering (n = 21), psychology (n = 1), collaborations between computer science and psychology (n = 8), and other disciplines (n = 4). The majority of the studies were conducted in laboratory environments, with only a few (n = 10) involving natural or field conditions.

3. Results

3.1. Data used in the studies

The data used in the reviewed studies can be broadly categorized into two types: publicly available datasets and self-collected datasets. Below, we examine their respective signal types, devices, collection protocols, and labelling strategies.

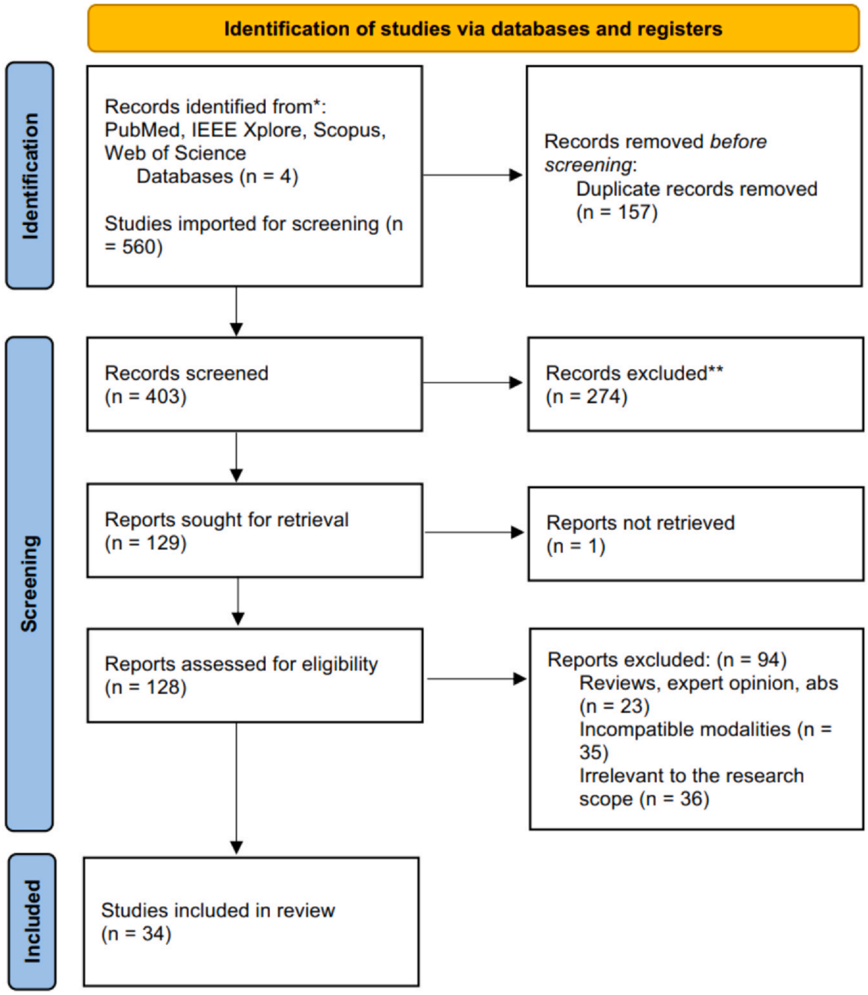

Fig. 1. PRISMA diagram of the literature review process.

3.1.1. *Public datasets*

Thirty studies used at least one publicly available dataset. The DEAP dataset [22], the DAPPER dataset [23], and the WESAD dataset [24] are the most commonly used resources. Specifically, the DEAP dataset contains peripheral physiological signals of 32 participants (50 % females, mean age 26.9) as each watched 40 one-minute long excerpts of music videos. The DAPPER dataset consists of wrist-worn physiological signals collected from 88 participants over five days without a stimulus. The WESAD dataset includes multimodal data collected from 15 participants (mean age of 27.5 ± 2.4 years) wearing devices on both their chest and wrist during standardized laboratory protocols. Detailed information on the public datasets is shown in Table 1.

Physiological signals span multiple modalities, including cardiac, electrodermal, respiratory, and motion-related signals. For clarity and consistency, signal modality names were standardized across studies. For example, galvanic skin response (GSR) and electrodermal activity (EDA) were uniformly referred to as EDA, as they represent the same underlying signal, despite the variation in terminology. The most commonly used modalities are EDA (n = 10/12), ECG (n = 7/12), and SKT (n = 5/12). A few studies have introduced other sensing data, such as SBP. In addition, the WESAD, DEAP, CASE, POPANE, and Emognition datasets provide rich multimodal signals (≥ 5 types). Nearly all the studies involved cardiac activity signals.

Most studies use diverse devices to measure multimodal signals collaboratively and align them on a timeline to ensure that subsequent data from each modality can be fused for analysis. Wrist-worn devices, such as the Empatica E4 and hand-worn devices, are widely used to capture signals such as PPG, ACC, and SKT. Chest-worn systems such as RespiBAN are employed primarily for acquiring signals such as ECG and RSP. Adhesive biomedical patches are widely applied to body locations such as the finger, upper arm, or back for measuring EMG and RSP. The reviewed datasets utilize both research-grade sensors (3 out of 12) and commercial wearable devices (9 out of 12).

The majority of public datasets were collected in controlled laboratory environments, with approximately 11 of the 12 datasets incorporating emotion elicitation tasks. These tasks typically adopted established emotion elicitation paradigms. Audiovisual or pictorial stimuli, such as film segments (e.g., MAHNOB-HCI), music videos (e.g., DEAP), or emotionally annotated pictures (e.g., GAPED), were presented to induce specific affective states under standardized procedures. Several datasets did not rely on predefined emotional stimuli but instead collected signals in simulation contexts or everyday real-life scenarios. For instance, K-EmoCon captured affective responses during natural debate interactions. DAPPER collected signals in real-world, free-living conditions, without any structured induction procedure. These settings aim to improve ecological validity, but they introduce additional challenges in labelling and modelling because of the spontaneous and context-dependent nature of emotional expression.

The datasets also collect subjects' actual affective states, which serve as ground-truth labels for model training and evaluation. Labelling strategies varied substantially depending on the data collection context. In laboratory settings with predefined emotional stimuli, self-reports were typically obtained immediately after each task using standardized instruments such as the SAM (Self-Assessment Manikin, n = 8/12) or

Table 1
Overview of the core features of public datasets.

Dataset	Stimuli	Subject number	Signal modality	Label Acquisition	Device	Used by
AMIGOS [25]	Video	40	ECGa, EDAb	SAMc	Shimmer 2R	[26–28]
ASCERTAIN [29]	Video	30	ECG, EDA	SAM	Commercial device (brand not mentioned)	[27,30]
CASE [31]	Video	30	ECG, EDA, SKTd, PPGe, RSPf, BVPg, EMGh	SAM	Lab sensors (hand and body)	[32–35]
DAPPER [23]	None	88	ACCi, PPG, EDA, HRj	ESMk, PANASl	Huixin wristband	[36–38]
DEAP [22]	Video	22	ECG, PPG, EDA, RSP, SKT	SAM	Lab sensors (hand)	[39–42]
Emognition [43]	Film	43	HR, BVP, ACC, GYROm, ROTn	SAM	Empatica E4, Samsung Galaxy SM-R810	[44]
GAPED [45]	Picture	13	EDA, PPG	SAM	Shimmer3	[46]
Juancq [47]	Film and music	50	ACC, GYRO, HR	PANAS	Polar M400, Samsung Gear 2, Polar H7	[48]
K-EmoCon [49]	Debate simulation	32	BVP, EDA, HR, SKT	Self-, partner-, and observer-annotated	Empatica E4, Polar H7	[30,50,51]
MAHNOB-HCI [52]	Video	27	ECG, EDA	SAM	Lab Sensors (hand and body)	[27]
POPANE [53]	Picture, film	1157	ECG, EDA, SBPo, DBPp, SKT	Expert-annotated	VU-AMS	[54]
WESAD [24]	Video	15	ECG, EDA, RSP, ACC, BVP, SKT, EMG	SAM, PANAS	RespiBAN, Empatica E4	[32,33,55–58–61]

[a] Electrocardiogram
[b] Electrodermal Activity
[c] Self-Assessment Manikin
[d] Skin Temperature
[e] Photoplethysmography
[f] R0espiration
[g] Blood Volume Pulse
[h] Electromyography
[i] Accelerometer
[j] Heart Rate
[k] Experience Sampling Method
[l] Positive and Negative Affect Schedule
[m] Pulse-to-Pulse Interval
[n] Gyroscope
[o] Rotation
[p] Systolic Blood Pressure [17]Diastolic Blood Pressure

ratings by subjects and observers (n = 2/12). Studies conducted in real-world or semistructured environments often employ experience sampling methods (n = 1/12) or questionnaires such as PANAS (Positive and Negative Affect Schedule, n = 3/12) to record emotions as they occur or shortly after.

The number of subjects ranges from small-scale lab datasets with fewer than 20 participants (e.g., WESAD, GAPED) to large-scale population studies (e.g., POPANE with 1157 participants). Most datasets involved 20–50 participants (n = 8/12).

3.1.2. *Self-collected data*

Nine studies adopted self-collected data, and 3 studies combined both public and self-collected data for comparison. Table 2 summarizes the core features of the self-collected data.

The self-collected datasets covered a broad range of physiological signals, including EDA (n = 9/9), SKT (n = 5/9), HR (n = 4/9), and PPG (n = 3/9). Similar to public datasets, EDA, SKT, and HR remain the most frequently used signals. One self-collected dataset explored eye-tracer signals such as pupil diameter and saccadic amplitude, which are less commonly seen in public resources.

In addition to employing commercial wearable devices similar to those used in public datasets (e.g., Empatica E4), self-collected datasets adopted customized devices more frequently. Customized setups enabled researchers to integrate additional physiological sensors (e.g., blood oxygen and pupil dilation) and flexibly place them on targeted body sites such as the hand, chest, or head, thus facilitating the exploration of novel signal combinations and task-dependent sensing strategies. The length of the recordings varied across datasets, ranging from a few minutes to several hours.

Label acquisition approaches varied across studies. Three studies used SAM, while the others relied on retrospective surveys or self-reported ratings. None of the reviewed studies employed in-the-moment experience sampling.

Table 2
Overview of the core features of self-collected datasets.

Stimuli	Subject number	Signal modality	Label Acquisition	Device	Used by
Colour Word	98	PPG, EDA, RSP	SAM	Customized device (hand, chest)	[62]
Medical simulation*	10	ECG, EDA	Self-annotated	Shimmer3	[27]
Medical simulation*	201	EDA, HR, BVP, SKT, ACC	None	Empatica E4	[63]
Video	10	EDA, SKT, HR	SAM	Empatica E4	[64]
Video	30	PPG, EDA, SKT	SAM	Customized device (wrist)	[65]
Video	37	PPG, EDA	Self-annotated	Customized device (hand)	[66]
Video	20	SKT, EDA, HR, BVP, Pupil dilation	Self-annotated	Empatica E4, Pupil Core	[35]
Video, walking simulation*	30	ECG, EDA, SKT, BVP	Self-annotated	Customized device (wrist)	[67]
Walking simulation	40	EDA, ACC, HR	Retrospective survey	Empatica E4	[68]

* refers to datasets that were collected in previous studies and have not been publicly released.

Five studies employed video stimuli to elicit emotional responses. One study used a Stroop colour and word test to induce cognitive–affective load. Two datasets used medical stimuli in a recreated hospital environment. The number of subjects ranged from 10 to 201, with most studies (n = 7/9) involving fewer than 50 participants.

Both public datasets and self-collected data play important roles in affective computing studies. They show high consistency in terms of modality, device, signal length, number of subjects, and label acquisition. For example, they both make extensive use of common physiological modalities such as EDA and HR and rely mainly on commercially available wearable devices (e.g., Empatica E4) for their measurements, most of the labels are acquired by self-assessment tools (e.g., SAM scales), and the number of subjects is generally on the scale of tens of people. Notably, some of the self-collected data incorporate sports-related scenarios, such as walking simulations, which record multimodal signals (including EDA, ACC, HR, etc.) to capture affective changes in individuals during physical activity. These types of data hold potential value for applications in the sports field, such as monitoring emotional fatigue during training or assessing athletes' emotional regulation capabilities under competitive pressure.

Nonetheless, there are some specific differences between the two, such as the fact that public datasets usually cover a richer mix of modalities (e.g., ECG, EMG) and more standardized device configurations, whereas self-collected data explore more custom devices and novel modalities (e.g., eye movement data). However, these differences are relatively secondary to some of the shortcomings and challenges shared by both. For example, existing datasets lack continuous emotion monitoring over a long period (days to months), which makes it difficult to reveal long-term emotion patterns. The number of subjects is still insufficient, and there is a lack of large-scale population samples of thousands or even tens of thousands of subjects. Existing datasets seldom involve real-time labelling and are thus unable to accurately capture the subtle dynamics of emotional changes and contextual information of real-life scenarios. Achieving larger-scale data collection, long-term continuous monitoring, and real-time label acquisition is essential for advancing affective computing towards more robust and ecologically valid applications.

3.2. Processing and fusion of multimodal physiological signals

The processing flow of wearable multimodal physiological signals usually consists of three key stages: signal preprocessing, feature extraction, and multimodal fusion. Although there are variations in specific approaches across studies, most of the work follows this generalized framework. The following section summarizes and analyses the representative methods in these three stages to reveal the mainstream technical paths and core challenges in current studies.

3.2.1. Signal preprocessing

Preprocessing of wearable multimodal data must address two major challenges: (1) handling the complex and variable noise inherent in wearable signals and (2) coordinating multimodalities with different sampling rates and recording mechanisms.

Addressing the first challenge requires dealing with noise. A key characteristic of wearable physiological data is their susceptibility to complex and variable noise. In comparison, the noise in nonwearable laboratory physiological signals (e.g., EEG) is relatively stable, allowing for the optimization and fixation of filtering parameters and thresholds [69]. These parameters, however, are often unsuitable for wearable data, where noise characteristics vary greatly across conditions and subjects. Another characteristic of wearable signals is the variation in both temporal resolution and recording duration. The resolution ranges from high-frequency signals, such as ECG, to low-frequency measurements, such as RSP. The recording duration ranges from short sessions of only a few minutes to continuous measurements over several days, exhibiting pronounced nonstationarity and dynamic shifts in affective states. Furthermore, labelling strategies also vary substantially. While laboratory settings allow for relatively precise temporal labelling aligned with predefined stimuli, daily life data are subjective and time-delayed through experience sampling or review questionnaires. These characteristics underscore the need for more robust and adaptive preprocessing of wearable data.

The second challenge lies in coordinating multimodalities. Different sensors often have distinct sampling rates and recording mechanisms, making direct alignment difficult. Without proper synchronization, multimodal information can become inconsistent, limiting the effectiveness of fusion and downstream modelling. To address both challenges, most studies adopt a multistep preprocessing pipeline that typically involves time alignment and resampling, filtering and denoising, normalization, and signal segmentation and selection.

First, time alignment and resampling can ensure that the sampling frequencies and timestamps of different modalities are unified. Common alignment strategies include downsampling high-frequency signals to match low-frequency modalities and periodically calibrating device clocks to the correct timestamp drift, hence ensuring that signals from different sensors are temporally synchronized. Common interpolation techniques involve identifying local anomalies based on statistical characteristics (e.g., mean, standard deviation, or median) within a sliding window [55] and filling missing or corrupted segments using linear interpolation, spline interpolation, or neighbour-based averaging, thus preserving the continuity and structural integrity of the physiological time series [50,63].

Second, the data are filtered and denoised. Common sources of wearable signal interference include motion artefacts caused by individual actions, cross-talk between physiological signals, ambient electromagnetic noise, and poor sensor–skin contact [70]. Classical filtering techniques remain widely used. High-pass filters are commonly used to remove low-frequency drift or baseline fluctuations, low-pass filters effectively attenuate high-frequency noise (such as motion-induced distortions) [27,39,60,68], and bandpass filters focus on specific frequency ranges to extract physiologically meaningful rhythmic features. In addition, smoothing techniques such as moving average filters and Savitzky–Golay filters are frequently used to reduce transient noise and emphasize low-frequency trends within the signal [56]. For wearable signals with highly dynamic and heterogeneous noise, determining appropriate filter parameters (e.g., cut-off frequencies) is challenging. Traditional approaches often require manually tuned filter cascades to handle different noise components, which limits their adaptability in real-world scenarios [71]. To overcome this limitation, adaptive filtering techniques have been introduced in 2 studies to automatically adjust filter coefficients in response to changing signal and noise characteristics. For instance, Wu et al. [61] used a least mean-squares (LMS) adaptive algorithm to minimize the error to the desired filter impulse response coefficients. Pradhan et al. [57] utilized an augmented unscented Kalman filter (AUKF) to eliminate interference problems and preserve the signals' local information automatically. These adaptive filtering methods reduce the reliance on predefined thresholds and improve noise suppression under dynamic conditions. Apart from filtering methods, wearable data preprocessing has increasingly adopted signal decomposition and source separation techniques to address noise issues. For example, the wavelet transform is a powerful multiscale tool for nonstationary signal analysis, decomposing input signals into scale-dependent coefficients to extract transient changes and localized events [57,59].

Third, normalization is used to remove interdevice, intermodal, or interindividual differences [51]. Common strategies include min–max scaling, z score standardization, and baseline correction, which ensure that signals from different sensors and subjects are comparable. Finally, window slicing and selection are used to divide the long time series signal into fixed-length windows, while segments severely affected by artefacts or other irrecoverable distortions are excluded to ensure data quality for subsequent analysis.

3.2.2. Feature extraction

In multimodal wearable emotion recognition studies, feature extraction strategies can be broadly categorized into two types: traditional handcrafted feature extraction methods and automatic feature learning methods.

Manual feature extraction extracts physiological features from the raw signal based on domain knowledge. For example, ECG and PPG signals are commonly used to extract features such as the SDNN and RMSSD [61,68]. EDA signals are used to derive features such as SCL and SCR [27,60]. SKT signals are used to extract features such as the mean temperature. These features are typically extracted within a sliding time window and can be grouped into three categories: time-domain features (e.g., mean, standard deviation, maximum and minimum values), frequency-domain features (e.g., power spectral density and band-specific energy), and nonlinear metrics (e.g., entropy measures and fractal dimension).

Overall, manual feature extraction focuses on deriving statistical indices from the original data. When dealing with large-scale, high-dimensional, and long-duration continuous wearable data, such approaches may face limitations in computational efficiency and capturing dynamic signal characteristics. In recent years, automatic feature extraction methods have been increasingly adopted.

Automatic feature extraction methods typically use machine learning and deep learning models to learn high-dimensional representations from preprocessed multimodal signals. These methods do not require predefined metrics such as the SDNN and mean temperature but instead extract vectorized features by automatically capturing waveform structures, rhythmic variations, and temporal patterns through operations such as convolution and attention mechanisms. For example, Pradhan et al. [56] used the ResNet 152 and InceptionV3 architectures to extract features from processed signals. Pallapothula et al. [33] utilized a temporal convolutional network to give each normalized signal its coded representation. These methods are well suited for scenarios that require modelling long time sequences, as the models can capture long-range temporal dependencies such as autocorrelation over extended lags and key dynamics within the signals [72,73]. These temporal characteristics are crucial for affective computing, where emotional states evolve gradually and physiological responses often exhibit delayed effects [74].

In recent years, the application of pretraining strategies in wearable emotion recognition has been increasing. By prelearning general physiological feature representations on large-scale, unlabelled or weakly labelled data, the model's transferability in downstream tasks can be significantly improved. Wu et al. [63] used self-supervised pretraining to extract generalized feature representations in simulation scenarios such as doctor consultation and diagnostic announcement. The pretrained model achieved an accuracy of 84.81 % in the valence classification task, compared with 81.51 % without pretraining, demonstrating the substantial benefit of incorporating pretraining strategies for downstream affective recognition. Despite their advantages in handling large-scale and long-duration data, automatic feature extraction still presents certain challenges. On the one hand, the learned features often lack clear physiological interpretability and are difficult to associate directly with specific physiological mechanisms. On the other hand, when signal quality is poor, this method is prone to introducing noisy or misleading features [75].

Notably, some studies have attempted to combine the strengths of both manual and automatic feature extraction. For example, EDA and PPG signals were transformed into time-frequency spectrograms and then input into CNNs to extract features [55]. Such integration may offer a balance between physiological interpretability and computational efficiency.

In summary, current studies adopt a range of feature extraction strategies, from physiologically grounded manual methods to data-driven automatic approaches, with some studies exploring combinations of the two. The choice between these strategies often depends on the specific application context, data characteristics, and modelling goals.

3.2.3. Fusion strategies

There are three fusion strategies: feature-level fusion (integrating modality-specific features during preprocessing), model-level fusion (combining representations during model training), and decision-level fusion (aggregating outputs at the decision stage). As illustrated in Fig. 2, these strategies differ in the timing and manner of information integration, with each offering distinct advantages for affective computing tasks. The feature fusion strategies used in the reviewed studies are shown in Table 3.

Feature-level fusion integrates multimodal information by concatenating features from different modalities into a unified input before model training. For instance, Ross et al. [27] implemented this approach by combining hand-crafted features from EDA and ECG signals, followed by unsupervised representation learning. Their results demonstrated that feature-level fusion achieved a classification accuracy of 64 %, outperforming decision-level fusion (59 %) under the same conditions. This approach highlights the advantage of early integration in capturing information across modalities, providing a foundation for models to learn latent interactions and cross-modal correlations.

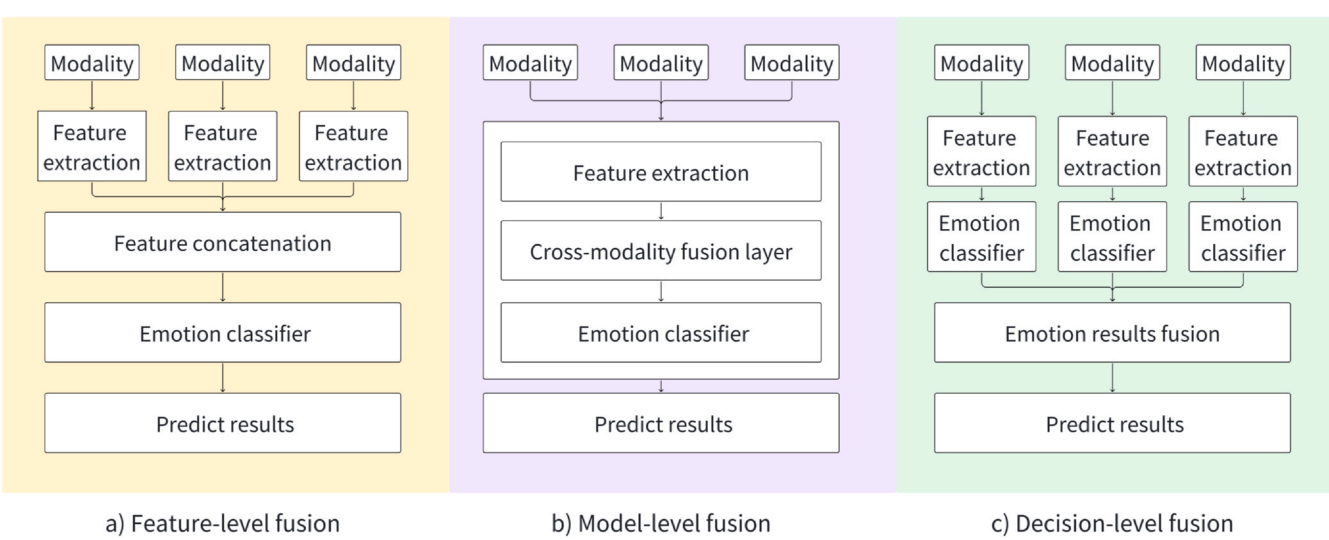

Fig. 2. Comparison of the three multimodal fusion strategies. a) Feature-level fusion, b) Model-level fusion, c) Decision-level fusion.

Table 3
Overview of the studies.

Procedure	Approaches	Used by
Feature fusion strategy	Feature-level	[27,32,33,37–40,41,42,44,48,50,51,54–57,61,62,68]
	Model-level	[30,34,36,58,60,63,67,76]
	Decision-level	[26,59,65]
Model	Machine learning model	[38,39,42,46,48,50]
	Deep learning model	[27,30,33,34,36,51,55,58,63,65,67,68,76]
Platform	Matlab	[46,50,51,54,55,59,62,64,68]
	Python	[27,39,40,51,55–57,59,60,62,76]

However, feature-level fusion also presents several challenges. It increases feature dimensionality, which may lead to higher computational costs and overfitting. Moreover, it requires precise temporal synchronization across modalities. Missing or misaligned data can introduce noise and degrade model performance.

Model-level fusion integrates multimodal signals at an intermediate layer of the model or through specialized mechanisms. For example, Paz-Arbaizar et al. [76] modelled cross-modal interactions using a shared transformer encoder. Wu et al. [63] connected multimodal signals in tandem through an attention mechanism layer for model training, achieving an emotion classification accuracy of 77.49 %, which outperforms the 73.01 % achieved using feature-level fusion. This technique simulates the heterogeneity and coordination of multimodal physiological signals simultaneously, which partially alleviates the requirements of data synchronization and captures the complementary information between modalities. Model-level fusion requires the design of complex model architectures (e.g., multilayer networks or cross-modal attention modules), which may lead to overfitting risks or longer training times.

Although multimodal fusion has become an important direction in affective computing, related studies have gradually increased since 2021 and are still in the early stages of development. Current mainstream methods often align all modalities temporally, ignoring potential information redundancy and interference issues between modalities. Additionally, different physiological signals originate from distinct regulatory systems, resulting in fundamental differences in physiological mechanisms, noise characteristics, and response patterns [77]. Simple concatenation may obscure critical dynamic features within the signals. For example, HR responses typically exhibit a delay of several seconds, whereas signals such as EMG can immediately reflect emotional activation. This difference in response latency has not been adequately addressed in existing fusion methods [78]. These limitations suggest that fusion approaches should consider the temporal alignment and physiological relevance of each modality.

Decision-level fusion means that the modalities first train their respective sentiment classifiers separately, and fusion occurs at the decision output stage. For example, Yan et al. [59] equip each modality with a subclassifier, which is fused by assigning different weights. This approach achieves a sentiment classification accuracy of 87.41 %, which is significantly higher than that of the majority voting-based fusion method, which is 82.22 %. Decision-level fusion allows the modal subsystems to remain relatively independent. If the signal quality of one sensor is poor, the classification results of other modalities can still provide reliable support. In addition, each modality can adopt its own data processing and modelling methods. The disadvantage of this approach is that decision-level fusion does not fully utilize the complementary information between modalities and cannot effectively capture the correlation of multimodal features.

Most studies adopt a feature-level fusion strategy, which may be because multimodal signals collected in controlled environments exhibit high synchrony and low noise, making feature-level concatenation and modelling more convenient. As research increasingly shifts towards daily life scenarios, issues such as signal asynchrony, missing data, and noise become more prominent. Model-level and decision-level fusion are receiving increased attention, as they learn latent relationships across modalities to enhance robustness and adaptability. Overall, the choice of fusion strategy is jointly influenced by dataset properties and application requirements. Hence, this area requires further systematic investigation in the future.

In summary, the reviewed studies indicate that multimodal fusion in affective computing can be implemented at different stages of the modelling pipeline, including feature-level, model-level, and decision-level fusion. Feature-level fusion provides a simple and easy-to-operate mechanism, model-level fusion can capture cross-modal interactions within the network structure, and decision-level fusion allows for independent processing of each modality. The selection of fusion strategies typically depends on the temporal characteristics, complementarity, and reliability of the involved modalities, as well as the complexity of the classification task.

3.3. Modelling Approaches and Performance Analysis

We summarize the trends in the application of machine learning and deep learning models in multimodal affective computing via wearable devices and then further explore the strengths and limitations of current models.

3.3.1. Model Approaches

Modelling approaches can be broadly categorized into conventional machine learning algorithms (such as support vector machines and random forests) and deep learning architectures, specifically approaches based on neural network architectures (e.g., CNN, LSTM, and transformer), as detailed in Table 3. The classical model usage frequency has evolved over time in the field of multimodal emotion recognition. In the following, we analyse different model types and their improvement strategies, as well as performance evaluation and data partitioning strategies, and summarize the current status and development trends of current model applications.

Machine learning approaches are usually trained on handcrafted multimodal features based on statistical rules or methods. Representative algorithms include decision trees, support vector machines (SVMs), random forests (RFs), and logistic regression (LR). These models are lightweight, require few computational resources, and are relatively easy to implement. For example, Pollreisz et al. [64] achieved a 4-class classification accuracy of 64.66 % using a decision tree with a small footprint and relaxed resource requirements. Another representative approach is the SVM. Ayata et al. [42] used an SVM to construct optimal classification boundaries by mapping the input features to a high-dimensional space through a radial basis kernel (RBF), which achieved an accuracy of 72.18 %. Ahmed et al. [38] further compared the emotion recognition accuracy of SVM and RF, and the latter model achieved a classification accuracy of 64 % through majority voting from decision trees. Overall, machine learning methods have significant advantages in terms of computational efficiency and interpretability of results but still face challenges when dealing with complex, high-dimensional, or modally heterogeneous data.

Fig. 3

Deep learning methods, mainly convolutional neural networks (CNNs), recurrent neural networks (RNNs), transformers, and hybrid

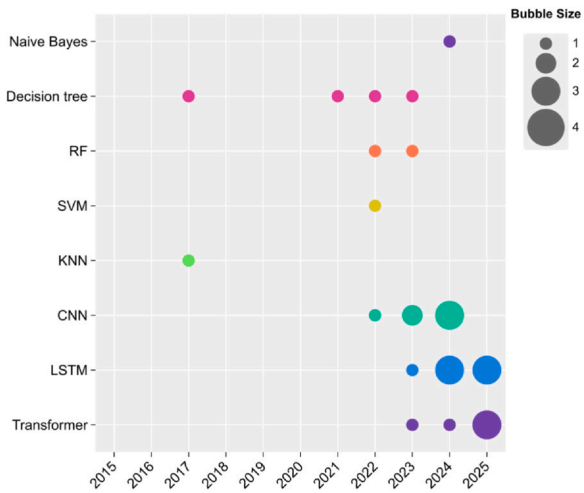

Fig. 3. Trends of Model Usage in Multimodal Emotion Recognition. The x-axis represents publication years, while the bubble size corresponds to the number of studies employing a given model.

structures, have been widely used in emotion recognition tasks in recent years. The hierarchical architecture of CNNs enables them to learn hierarchical feature representations from original signals, and they are widely used to transform physiological signals into spectral features to capture local physiological changes. For example, Wu et al. [61] achieved a 4-class classification accuracy of 87.90 % using a CNN. However, the localized convolutional kernel structure of CNNs limits their ability to perceive global physiological patterns over long time windows. This limitation poses a challenge because changes in affective states tend to exhibit long-term temporal features. Recurrent neural networks (RNNs), especially long short-term memory (LSTM), effectively capture local temporal dependencies in temporal data through their recursive structure and are suitable for addressing time series variations in physiological signals. Singh et al. [44] achieved a 9-class classification accuracy of 96.82 % by combining CNNs for spatial feature extraction with LSTM networks for modelling temporal dependencies. These architectures still face the problem of vanishing or exploding gradients in long time series. Transformers overcome these limitations through the self-attention mechanism and are able to directly capture long-range temporal dependencies, making them suitable for processing complex and prolonged physiological signal segments. Some studies have tried to apply this approach in affective computing tasks. Wu et al. [63] achieved an accuracy of 81.51 % for 3-class emotion recognition using a transformer model, compared with 80.53 % without the transformer. This model can learn the dynamic relationships of features in a global context and especially excels in the processing of long, complex signals. However, challenges remain in using the transformer with a small sample or under highly noisy data conditions. In these contexts, the model is prone to overfitting, leading to performance degradation. To address these issues, some studies have begun to explore self-supervised learning methods to improve the performance of models in data-scarce or noisy situations. For example, Wu et al. [63] improved the classification accuracy by 5.24–13.63 % through self-supervised learning of a fine-tuned model with parameter initialization.

Studies modifying learning models can be categorized into three primary types: parameter tuning based on existing frameworks, customized deep learning architectures, and model lightweighting. Parameter tuning adjusts the parameters based on existing frameworks. For example, Pollreisz et al. [64] used a model in [79] by analysing the peaks in signals. Customized deep learning architectures enhance the performance by designing specific network structures, such as by combining a transformer with a GRU to process cross-modal data. Model lightweighting focuses on resource efficiency optimization. For example, Chen et al. [41] used pruning and quantization techniques to make a deep model suitable for resource-constrained devices such as wearable sensors or mobile phones.

Emotion recognition tasks can be classified into two main categories: 1) Discrete emotion classification tasks, where emotions are categorized into distinct classes based on the basic emotion theories [80] or are extended to broader psychological states (e.g., stress, anxiety, immersion, heart flow, etc.) [81,82]. These tasks use physiological signals to predict specific labels through learning models. 2) Continuous dimensional emotion classification tasks, which use the valence–arousal model to predict individual emotional states [83].

The selection of baseline models is based mostly on traditional machine learning algorithms or existing unimodal and multimodal emotion recognition frameworks. Many studies use CNNs as controls (used in 18 studies). For instance, Ahmed et al. [38][37] used classical architectures (AlexNet, VGG, and other CNNs) for comparison. Elalamy et al. [26] used different deep learning network structures, such as AlexNet and CNN, as baseline models.

In terms of performance metrics, accuracy, precision, recall, F1 value, ROC, and AUC are commonly used in various studies for evaluation. Moreover, the data partitioning strategy directly affects the reliability and reproducibility of the model evaluation. We have systematically summarized the training: testing: validation data division ratios in the literature. The most common division ratio is 80:0:20, followed by 70:0:30, 70:20:10, and 75:15:10, and some studies also use the more refined hold-out method. Most of the validation work uses classical ratio division or cross-validation, whereas individual studies use special division methods such as leave-one-subject-out or nested k-fold.

The model architecture and implementation methods used in the reviewed studies involve different computing platforms and tools. Eight studies used MATLAB, two studies used MATLAB and Python, and the remaining 24 studies used mainly the Python TensorFlow and PyTorch platforms. Many studies did not disclose their complete code. Only the EmoHEAL [30] framework provided GitHub code. Other studies rarely mentioned whether they were open source, making it challenging for other researchers to reproduce these results.

We explore modelling approaches in multimodal affective computing, covering machine learning and deep learning techniques in recent years. In machine learning, which is used in approaches such as decision trees and SVMs, training is conducted through the use of statistical rules and handcrafted features. This method has the advantage of low resource requirements but is limited in processing complex and high-dimensional data. Deep learning methods extract and model feature representations of input data. Common models include CNNs for extracting local features, LSTM for capturing temporal dependencies, and transformers for addressing temporal dependencies over long distances through a self-attention mechanism. We further analyse the different optimization strategies of these models, the classification of emotion recognition tasks, the selection of performance evaluation metrics, and the impact of data partitioning strategies on model reliability.

3.3.2. Performance analysis of unimodal and multimodal models

This section analyses the performance of emotion recognition models. It focuses on comparing the effects of multimodal and unimodal approaches in different datasets and task settings, and explores the performance differences between different modelling approaches. Through systematic comparisons, we summarize the performance of each model in terms of accuracy and other metrics and analyse the impact of the modality and model architecture on the results.

Before analysing the performance of the multimodal models, we examined the frequency of each signal modality across all the reviewed studies. Fig. 4 presents the usage frequency of each modality. EDA and ECG signals are commonly used measurement modalities, followed by SKT and HR. Other modalities, such as BVP, RSP, and EMG, are used

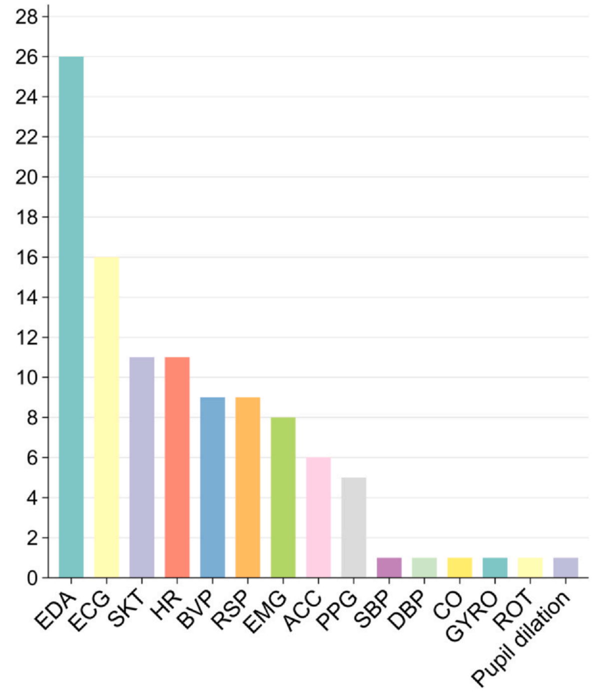

Fig. 4. Frequency distribution of modalities used in studies.

with moderate frequency. A few studies use modalities such as CO, GYRO, and ROT.

Afterwards, we compared the performance of multimodal versus unimodal approaches across multiple datasets and task settings, as shown in Table 4. The results show that multimodal models outperform the best unimodal models based on the same architecture and parameters, primarily because of the effective integration of complementary information between modalities. For instance, Alskafi et al. [50] achieved a maximum accuracy of close to 83 % when 4 modalities were used and 70 % when 3 modalities were used, which is better than the 51 % accuracy achieved when the single modality SKT was used, thus showing a significant enhancement of multimodal fusion in emotion recognition tasks. Multimodal signals may be used to enhance the accuracy of emotion classification because affective responses involve multiple physiological systems [78]. These systems include the cardiovascular system, respiratory system, autonomic nervous system (ANS), muscular system, and thermoregulatory system. These systems respond to external stimuli through complex physiological feedback mechanisms and exhibit dynamic features in their signals [78]. For example, the cardiovascular system, which is regulated by the ANS, produces changes in HR, which are visible in ECG waveforms and pulse signals from BVP. The respiratory system is affected by emotions, and the respiratory rate increases during tension or excitement, which can be captured by RSP signals. The electrical response of the skin caused by the secretion of sweat glands is reflected in EDA, and muscular tension may interfere with other signals. The multimodal approach can simultaneously integrate complementary information between these systems, resulting in enhanced expressive and generalization abilities in complex affective computing tasks. The enhancement of multimodal models in some studies is small. For example, Gohumpu et al. [39] fused HR, SKT, and EDA signals, which resulted in a 0.007 increase in accuracy relative to the SKT modality. This result suggests that multimodal modelling is not simply a matter of adding modal information together. Instead, the characteristics of the dataset, the complementarities between modalities, and the fusion strategies should be considered.

In this section, the performance of emotion recognition models is analysed, with a focus on comparing the performance of multimodal and unimodal methods across different datasets and task settings. Overall, multimodal models outperform the best unimodal models in most studies.

3.3.3. *Performance optimization strategies*

The performance improvement in emotion recognition tasks is related not only to the multimodal signals and the way they are fused but also to the model architecture. Model architecture has always been a central component in related studies, and researchers have proposed a variety of methods to optimize the performance of models. Based on the magnitude of changes to the model, we classify the model optimization methods covered in the studies into three main categories: parameter tuning based on existing frameworks, model reconstruction according to task requirements, and model lightweighting. Table 5 shows the performances of the proposed models and the corresponding best baseline models in the reviewed studies.

Parameter tuning improves the performance of an existing model by adjusting its hyperparameters without changing the original model architecture. This method focuses on finding the best parameter configurations to improve the model performance without changing the original model architecture. Common tuning methods include choosing appropriate kernel functions and adjusting regularization parameters (e.g., C and gamma in the SVM), using different input feature vectors, adjusting hyperparameters such as the learning rate and batch size, etc.,

Table 4
Performance Comparison Between Multimodal and Unimodal Models.

Dataset	Task	Multimodal Acc	Unimodal ACC	Multimodalities	Single Modality	Ref.
CASE	Emotion – 4	0.8291	0.7937	EDA, ECG	EDA	[32]
CASE	Arousal – 3	0.8669	0.8291	EDA, SKT, BVP	SKT	[33]
DAPPER	Valence – 2	0.8275	0.8174	PPG, EDA, ACC	PPG	[37]
DAPPER	Valence – 5	0.6026	0.4938	HR, EDA, ACC	HR	[36]
DEAP	Arousal – 2	0.7308	0.7132	RSP, PPG, SKT	RSP	[42]
DEAP	Valence – 2	0.6600	0.6530	HR, EDA, SKT	HR	[39]
WESAD	Emotion – 3	0.8000	0.7700	RSP, EDA, ACC, EMG, ECG	RSP	[60]
WESAD	Emotion – 3	0.8381	0.7872	EDA, BVP, SKT	EDA	[63]
WESAD	Emotion – 3	0.8388	0.7659	RSP, EDA, EMG, ECG	ECG	[58]
WESAD	Emotion – 3	0.8741	0.8515	RSP, EDA, EMG, ECG	ECG	[59]
AMIGOS	Arousal – 2	0.9300	0.8500	ECG, EDA	ECG	[27]
CVDiMo	Emotion – 6	0.9300	0.8100	ECG, EDA	ECG	[67]
K-EmoCom	Emotion – 4	0.8300	~ 0.5100	BVP, EDA, SKT, HR	SKT	[50]
K-EmoCom	Valence – 2	0.9291	0.7575	HR, EDA, SKT	EDA	[51]
Self-collected	Emotion – 3	0.7092	0.5497	EDA, HR, ACC	EDA	[68]
Self-collected	Emotion – 6	0.6790	0.4000	EDA, SKT	EDA	[65]

Emotion-4, Valence-2, and Arousal-3 denote the classification task and the number of target categories. “ ~ ” represents the value derived from the data shown in the figure.

Table 5
Performance comparison of the proposed model and the baseline model.

Model Innovation	Dataset	Task	Proposed ACC	Baseline ACC	Proposed Model	Baseline Model	Ref.
Parameter tuning	WESAD	Emotion – 3	0.8100	0.7900	AdaBoost decision trees	Gradient tree boosting	[60]
	Self-collected	Emotion – 3	0.6466	0.9000	Decision tree	SVM + GA	[64]
	Self-collected	Emotion – 3	0.7002	0.6145	Bagged trees	Neural Networks	[68]
Model reconstruction	CvdiMo	Emotion – 6	0.9300	0.7600	CNN-RF	CNN	[67]
	CASE	Valence – 2	0.7857	0.7329	TCN + Transformer	SigRep	[63]
	DAPPER	Valence – 5	0.6155	0.5732	Transformer	RNN	[36]
	K-EmoCon	Valence – 2	0.9334	0.7875	ER	XGBoost	[51]
	K-EmoCon	Arousal – 2	0.7296	0.6977	TCN + GRU	ResNet + GRU Fusion	[30]
	MAHNOB-HCI	Arousal – 2	0.8900	0.8500	Autoencoder	CNN	[27]
	WESAD	Emotion – 3	0.8500	0.7251	CNN LSTM Attention	ABDT	[58]
	WESAD	Emotion – 3	0.9524	0.9205	RFDNN	RNN	[55]
	WESAD	Arousal – 3	0.8669	0.8538	GM-FuseNet	SSL [85]	[56]
	WESAD	Emotion – 3	0.6654	0.6435	HEPLM	SVM	[33]
	WESAD	Emotion – 4	~0.8200	~0.5600	SE-CNN	LDA	[61]
	Self-collected	Emotion – 6	0.9560	0.9544	LMFN	[86]	[65]
Model light weighting	DEAP	Emotion – 6	0.8200	0.8050	ShuffleNetV2	LSTM	[41]
	Juancq	Emotion – 3	0.8940	0.8530	LightGBM	RF	[48]
	Emognition	Emotion – 9	0.9682	0.8530	Hybrid CNN-LSTM	RF	[44]

Emotion-4, Valence-2, and Arousal-3 denote the classification task and the number of target categories. "~" represents the value derived from the data shown in the figure.

or choosing different optimization algorithms (e.g., Adam and SGD) to improve the convergence speed and stability of the model. For example, Pollreisz et al. [64] used several existing algorithms (such as RF and SVM) on their dataset along with parameter tuning to improve the performance of emotion recognition models.

Although parameter tuning can improve the performance of a model in certain tasks, when the task becomes more complex, this method may not be able to fully exploit the multimodal data information. In this case, model reconstruction based on data characteristics and computational tasks tends to better meet the application needs, especially when dealing with high-dimensional and heterogeneous data. Common reconstruction methods include the following: 1) Layer adaptation: the number of layers in the network are adjusted, or certain layers are replaced with other types of layers. For example, Hssayeni et al. [60] adjusted the number of convolutional layers and modified the last fully connected layer to contain only one node with a linear activation function. The adjusted model achieved an accuracy of 79 % in the emotion recognition task, representing an improvement over the baseline model (accuracy rate of 72 %). 2) Module reorganization: different models or reusable modules are adjusted and combined within a model (such as encoders, decoders, etc.), enhancing the scalability and reusability of the model. For instance, Singh et al. [44] combined CNNs for spatial feature extraction with LSTM networks for modelling temporal dependencies and achieved an accuracy of 96.82 % on a 9-discrete-emotion emotions classification task. 3) The introduction of novel model structures: new architectural designs, such as transformers and graph neural networks (GNNs), are used. For example, when the transformer architecture was first proposed, it significantly improved performance in sequence-to-sequence tasks, showing particular advantages in handling long temporal dependencies [84]. In the field of affective computing, Wu et al. [63] used the transformer architecture for binary valence classification, achieving an accuracy of 77.49 %. When the transformer was not used, the accuracy was only 74.65 %. Overall, layer adjustment, module reorganization, and the introduction of new methods are commonly used model restructuring strategies. Layer adjustment improves the depth and complexity of the model by modifying the number of layers in the network or replacing layers. Module reorganization allows the model to be adjusted and optimized according to task requirements. Introducing new methods can solve problems that traditional models cannot handle well.

Some studies focus on model light weighting for resource-constrained scenarios. For example, Chen et al. [41] proposed ShuffleNetV2, which uses lightweight deep learning models that are specifically designed to operate efficiently on wearable sensors or mobile phones. The accuracy of ShuffleNetV2 ranges from 82 % to 86 %, achieving comparable accuracy to that of LSTM, which also falls within the 82–86 % range. These findings show that lightweight models achieve comparable accuracy to that of heavier models while significantly reducing computational overhead and power consumption. These lightweighting studies pave the way for the development of practical, real-time emotion recognition systems that can be deployed on a wide range of resource-constrained end devices.

This section summarizes the optimization strategies for the models. Common adjustment methods include parameter tuning, model reconstruction, and model light weighting. Through these adjustments, models can better adapt to the needs of different tasks and improve the accuracy of emotion recognition tasks.

4. Future directions

The application of wearable devices in the field of affective computing has made remarkable progress, yet several enhancements can still be made in the following areas.

First, future studies should prioritize the collection of large-scale, long-term, continuous wearable data in naturalistic settings. Although signals in these environments are subject to more complex noise, they more faithfully capture the dynamic evolution of real emotional states [87]. Evidence from recent studies has demonstrated that pretraining on large-scale continuous wearable datasets can substantially enhance the generalization of downstream models [63]. Expanding data acquisition via diverse wearable sensors in natural scenarios is therefore essential for capturing authentic emotional patterns and improving the ecological validity of emotion recognition [88–90].

Second, future studies should optimize multimodal fusion strategies and modelling approaches to better exploit the complementary information across modalities [91]. Existing methods still face challenges in addressing response asynchrony, feature fusion, and modality-specific noise, which constrain model performance in complex real-world environments. Future work could explore adaptive fusion architectures such as cross-modal transformers [92], as they are capable of jointly modelling spectral, spatial, and temporal patterns and leveraging the complementary information across modalities, and graph neural networks [93], which can capture intrinsic relationships across modalities and enable more comprehensive multimodal and contextual modelling. Moreover, self-supervised learning approaches, including contrastive learning frameworks [94] and masked autoencoders [95], make it possible to extract more robust and transferable feature representations by learning cross-modal consistency and temporal context dependence on large-scale, unlabelled multimodal data.

Finally, emphasis should be placed on the interpretability and lightweight design of the model to enhance its deployability and user trust in resource-constrained environments such as mobile devices and health monitoring, thereby truly realizing the everyday and inclusive application of affective computing technology [96,97].

Beyond general challenges, data collection and application in sports settings require particular attention. Future research should extend beyond collecting data during rest or daily activities to cover the entire athletic process systematically. During training or competition, multimodal signals can reflect athletes' tension, focus, and stress levels in real time, enabling them to adjust their affective state promptly to enhance performance [98]. For instance, Hongn et al. [99] recorded physiological signals during structured acute stress induction, as well as during aerobic and anaerobic exercise sessions. Such work provides an empirical foundation for future affective recognition and intervention tailored to athletic contexts. In the recovery phase after exercise, multimodal signals can reveal individual fatigue levels and affective state changes, providing coaches with personalized recovery and training recommendations [100]. Jiang et al. [101] analysed the patterns underlying the past kinematic sequence obtained from wearable sensors and generated fatigue progression predictions. By collecting multimodal signals before, during, and after exercise, we can comprehensively characterize athletes' affective and physiological states across different phases, thereby providing a basis for optimizing competitive performance and scientific training.

5. Conclusion

This review systematically analyses 34 studies on multimodal emotion recognition using wearable physiological signals over the past decade. Specifically, it provides a comprehensive overview of publicly available and self-collected datasets, summarizes the usage frequency and characteristics of various physiological modalities, and examines the role of signal processing workflows in enhancing data quality. The analysis revealed that EDA and ECG are the most frequently used modalities and that signal processing in wearable data needs to address issues such as complex noise, temporal misalignment across modalities, and the nonstationary dynamics of physiological signals to ensure high-quality data and robust model performance. The three main strategies for multimodal fusion are feature-level fusion, model-level fusion, and decision-level fusion. Notably, compared with unimodal methods, multimodal methods demonstrate superior classification performance in most studies, thereby validating the enhancing effect of cross-modal information integration on emotion recognition performance. Mainstream model architectures were categorized and compared, and their optimization trends were thoroughly explored. Commonly used models include classic machine learning methods (such as SVM and RF), as well as novel deep learning models (such as CNN, LSTM, and transformer). Moreover, compared with parameter adjustments, an increasing number of studies tend to reconstruct and optimize model structures to model the temporal characteristics and correlations in multimodal signals.

Despite remarkable progress, multimodal affective computing based on wearable devices still requires further exploration in several areas to enhance its ecological validity and practical application value. These areas include collecting large-scale natural data, incorporating athletic scenarios such as walking and running, advancing multimodal adaptive fusion strategies and modelling approaches, and developing lightweight models.

CRediT authorship contribution statement

Fang Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Dan Zhang:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization.

Declaration of Competing Interest

Dan Zhang is an Associate Editor for Intelligent Sports and Health and was not involved in the editorial review or the decision to publish this article. No financial relationships have influenced the research presented in this manuscript.

References

[1] R.W. Picard, Affective Computing, MIT Press, Cop, 1997.
[2] S. Afzal, H.A. Khan, Md.J. Piran, J.W. Lee, A comprehensive survey on affective computing: Challenges, trends, applications, and future directions, IEEE Access 12 (2024) 96150–96168.
[3] Y. Wang, W. Song, W. Tao, A. Liotta, D. Yang, X. Li, S. Gao, Y. Sun, W. Ge, W. Zhang, W. Zhang, A systematic review on affective computing: emotion models, databases, and recent advances, Inf. Fusion 83-84 (2022) 19–52.
[4] F.Z. Canal, T.R. Müller, J.C. Matias, G.G. Scotton, A.R. de Sa Junior, E. Pozzebon, A.C. Sobieranski, A survey on facial emotion recognition techniques: a state-of-the-art literature review, Inf. Sci. 582 (2022) 593–617.
[5] R. Guo, H. Guo, L. Wang, M. Chen, D. Yang, B. Li, Development and application of emotion recognition technology — a systematic literature review, BMC Psychol. 12 (2024).
[6] J. Kim, E. André, Emotion recognition based on physiological changes in music listening, IEEE Trans. Pattern Anal. Mach. Intell. 30 (12) (2008) 2067–2083.
[7] K.H. Kim, S.W. Bang, S.R. Kim, Emotion recognition system using short-term monitoring of physiological signals, Med. Biol. Eng. Comput. 42 (2004) 419–427.
[8] J. Chen, B. Xu, D. Zhang, Inter-brain coupling analysis reveals learning-related attention of primary school students, Educ. Technol. Res. Dev. 72 (2) (2024) 541–555.
[9] X. Shui, Y. Chen, X. Hu, F. Wang, D. Zhang, Personality in daily life: multi-situational physiological signals reflect big-five personality traits, IEEE J. Biomed. Health Inform. 27 (6) (2023) 2853–2863.
[10] Y. Liu, C. Jiang, Recognition of shooter's emotions under stress based on affective computing, IEEE Access 7 (2019) 62338–62343.
[11] L. Wang, Rapid recognition of athlete's anxiety emotion based on multimodal fusion, Int. J. Biom. 16 (5) (2024) 449–462.
[12] M. Gnanasigamani, E. Arslan, Y. Soylu, B. Kilit, P. Chmura, Mapping multi-modal fatigue in elite soccer through sweat-omics perspectives: a narrative review, Biology 14 (8) (2025) 1069.
[13] S. Saganowski, B. Perz, A.G. Polak, P. Kazienko, Emotion recognition for everyday life using physiological signals from wearables: a systematic literature review, IEEE Trans. Affect. Comput. 14 (3) (2023) 1876–1897.
[14] V.S. Machhi and A.M. Shah.A review of wearable devices for affective computing. In: 2024 International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC), 1-6, 2024.
[15] S. Ba, X. Hu, Measuring emotions in education using wearable devices: a systematic review, Comput. Educ. 200 (2023) 104797.
[16] B.A. Hickey, T. Chalmers, P. Newton, C.-T. Lin, D. Sibbritt, C.S. McLachlan, R. Clifton-Bligh, J. Morley, S. Lal, Smart devices and wearable technologies to detect and monitor mental health conditions and stress: a systematic review, Sensors 21 (2021) 3461.
[17] A.F. Claret, K.R. Casali, T.S. Cunha, M.C. Moraes, Automatic classification of emotions based on cardiac signals: a systematic literature review, Ann. Biomed. Eng. 51 (2023) 2393–2414.
[18] C. Anders, B. Arnrich, Wearable electroencephalography and multi-modal mental state classification: a systematic literature review, Comput. Biol. Med. 150 (2022) 106088.
[19] T. Röddiger, C. Clarke, H. Gellersen, P. Breitling, T. Schneegans, H. Zhao, and M. Beigl.Sensing with earables: A systematic literature review and taxonomy of phenomena.
[20] D. Moher, A. Liberati, J. Tetzlaff, D.G. Altman, Preferred reporting items for systematic reviews and meta-analyses: the prisma statement, BMJ 339 (2009) b2535.
[21] M.L. Hoang, et al., A review of developments and metrology in machine learning and deep learning for wearable iot devices, Comput. Sci. Eng. Med. (2025).
[22] S. Koelstra, C. Muhl, M. Soleymani, J.-S. Lee, A. Yazdani, T. Ebrahimi, T. Pun, A. Nijholt, I. Patras, Deap: a database for emotion analysis; using physiological signals, IEEE Trans. Affect. Comput. 3 (1) (2011) 18–31.
[23] X. Shui, M. Zhang, Z. Li, X. Hu, F. Wang, D. Zhang, A dataset of daily ambulatory psychological and physiological recording for emotion research, Sci. Data 8 (1) (2021) 161.
[24] P. Schmidt, A. Reiss, R. Duerichen, C. Marberger, and K. Van Laerhoven. Introducing wesad, a multimodal dataset for wearable stress and affect detection. In: Proceedings of the 20th ACM international conference on multimodal interaction, 400-408, 2018.
[25] J.A. Miranda-Correa, M.K. Abadi, N. Sebe, I. Patras, Amigos: a dataset for affect, personality and mood research on individuals and groups, IEEE Trans. Affect. Comput. 12 (2) (2018) 479–493.
[26] R. Elalamy, M. Fanourakis, G. Chanel, Multi-modal emotion recognition using recurrence plots and transfer learning on physiological signals, 2021 9th International Conference on Affective Computing and Intelligent Interaction (ACII), IEEE, Nara, Japan, 2021, pp. 1–7.
[27] K. Ross, P. Hungler, A. Etemad, Unsupervised multi-modal representation learning for affective computing with multi-corpus wearable data, J. Ambient Intell. Humaniz. Comput. 14 (4) (2023) 3199–3224.

[28] E.-J. Chang, A. Rahimi, L. Benini, A.-Y.A. Wu, Hyperdimensional computing-based multimodality emotion recognition with physiological signals, 2019 IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS), IEEE, 2019, pp. 137–141.
[29] R. Subramanian, J. Wache, M.K. Abadi, R.L. Vieriu, S. Winkler, N. Sebe, Ascertain: emotion and personality recognition using commercial sensors, IEEE Trans. Affect. Comput. 9 (2) (2016) 147–160.
[30] O. Tariq, Y. Oh, D. Han, and IEEE.EmoHEAL: A Fusion-Based Framework for Emotion Recognition Using Wearable Sensors.In: 2024 IEEE SENSORS, 2024.
[31] K. Sharma, C. Castellini, E.L. Van Den Broek, A. Albu-Schaeffer, F. Schwenker, A dataset of continuous affect annotations and physiological signals for emotion analysis, Sci. Data 6 (1) (2019) 196.
[32] P. SriramKumar, P.K. Govarthan, A. AleemShaikGadda, N. Ganapathy, J. FredoAgastinoseRonickom, Deep learning-based automated emotion recognition using multimodal physiological signals and time-frequency methods, IEEE Trans. Instrum. Meas. 73 (2024) 1–12.
[33] V.K.N. Pallapothula, S. Anand, S. Das Bhattacharjee, J. Yuan, Generalized multisensor wearable signal fusion for emotion recognition from noisy and incomplete data, Smart Health 36 (2025).
[34] N.T. Vu, V.T. Huynh, H.-J. Yang, S.-H. Kim, Multiscale Transformer-Based for Multimodal Affective States Estimation from Physiological Signals, in: H. Lu, M. Blumenstein, S.-B. Cho, C.-L. Liu, Y. Yagi, T. Kamiya (Eds.), Lect. Notes Comput. Sci., volume 14408 LNCS, Springer Science and Business Media Deutschland GmbH, 2023, pp. 113–122.
[35] T. Zhang, A. El Ali, C. Wang, A. Hanjalic, P. Cesar, Cornet: fine-grained emotion recognition for video watching using wearable physiological sensors, Sensors 21 (1) (2020) 52.
[36] F. Li, D. Zhang, Transformer-driven affective state recognition from wearable physiological data in everyday contexts, Sensors (Basel Switz.) 25 (3) (2025) 761.
[37] A. Ahmed, J. Ramesh, S. Ganguly, R. Aburukba, A. Sagahyroon, F. Aloul, Evaluating multimodal wearable sensors for quantifying affective states and depression with neural networks, IEEE Sens. J. 23 (19) (2023) 22788–22802.
[38] A. Ahmed, J. Ramesh, S. Ganguly, R. Aburukba, A. Sagahyroon, F. Aloul, Investigating the feasibility of assessing depression severity and valence-arousal with wearable sensors using discrete wavelet transforms and machine learning, Information 13 (9) (2022) 406.
[39] J. Gohumpu, M. Xue, Y. Bao, Emotion recognition with multi-modal peripheral physiological signals, Front. Comput. Sci. 5 (2023).
[40] A. Nandi, F. Xhafa, L. Subirats, S. Fort, Federated learning with exponentially weighted moving average for real-time emotion classification, in: V. Julián, J. Carneiro, R.S. Alonso, P. Chamoso, P. Novais (Eds.), Ambient Intelligence—Software and Applications—13th International Symposium on Ambient Intelligence, volume 603, Springer International Publishing, Cham, 2023, pp. 123–133.
[41] X. Chen, D. Liang, J. Zhang, Y. Zhang, Deploying Emotion Recognition on Resource Constrained Device: Utilizing Light Weight Deep Learning Models with Multi-Physiological Signals, Proc. Int. Conf. Signal Process., Comput. Networks Commun., SPCNC, Association for Computing Machinery, Inc, 2025, pp. 519–524.
[42] D. Ayata, Y. Yaslan, M.E. Kamasak, Emotion recognition from multimodal physiological signals for emotion aware healthcare systems, J. Med. Biol. Eng. 40 (2) (2020) 149–157.
[43] S. Saganowski, J. Komoszyńska, M. Behnke, B. Perz, D. Kunc, B. Klich, Ł.D. Kaczmarek, P. Kazienko, Emognition dataset: emotion recognition with self-reports, facial expressions, and physiology using wearables, Sci. Data 9 (1) (2022) 158.
[44] I. Singh, D. Nandini, J. Yadav, Integrated CNN-LSTM Model for Emotion Detection Using Physiological Signals from Wearables, in: S. Kumar, S. Hiranwal, R. Garg, S.D. Purohit (Eds.), Lect. Notes Networks Syst, 1122 Springer Science and Business Media Deutschland GmbH, 2025, pp. 363–374.
[45] E.S. Dan-Glauser, K.R. Scherer, The geneva affective picture database (gaped): a new 730-picture database focusing on valence and normative significance, Behav. Res. Methods 43 (2011) 468–477.
[46] G. Udovičić, J. Derek, M. Russo, M. Sikora, Wearable Emotion Recognition system based on GSR and PPG signals, MMHealth - Proc. Int. Workshop Multimed. Pers. Health Health Care, Co-Located MM, Association for Computing Machinery, Inc, 2017, pp. 53–59.
[47] J.C. Quiroz, E. Geangu, M.H. Yong, Emotion recognition using smart watch sensor data: Mixed-design study, JMIR Ment. Health 5 (3) (2018) e10153.
[48] Z.K. Nur, R. Wijaya, G.S. Wulandari, Gradient Boosting, AdaBoost, and LightGBM for Emotion Recognition: A Comparative Analysis on Wearable Sensor Data, Int. Conf. Inf. Commun. Technol., ICoICT, Institute of Electrical and Electronics Engineers Inc., 2024, pp. 585–591.
[49] C.Y. Park, N. Cha, S. Kang, A. Kim, A.H. Khandoker, L. Hadjileontiadis, A. Oh, Y. Jeong, U. Lee, K-emocon, a multimodal sensor dataset for continuous emotion recognition in naturalistic conversations, Sci. Data 7 (1) (2020) 293.
[50] F.A. Alskafi, A.H. Khandoker, U. Lee, C.Y. Park, H.F. Jelinek, Multimodal Analysis of Physiological Signals for Wearable-Based Emotion Recognition Using Machine Learning, 2022 Computing in Cardiology (CinC) 498 (2022), pp. 1–4.
[51] M.S. Zitouni, C.Y. Park, U. Lee, L.J. Hadjileontiadis, A. Khandoker, LSTM-modeling of emotion recognition using peripheral physiological signals in naturalistic conversations, IEEE J. Biomed. Health Inform. 27 (2) (2023) 912–923.
[52] M. Soleymani, J. Lichtenauer, T. Pun, M. Pantic, A multimodal database for affect recognition and implicit tagging, IEEE Trans. Affect. Comput. 3 (1) (2011) 42–55.
[53] M. Behnke, M. Buchwald, A. Bykowski, S. Kupiński, L.D. Kaczmarek, Psychophysiology of positive and negative emotions, dataset of 1157 cases and 8 biosignals, Sci. Data 9 (1) (2022) 10.
[54] X. Yang, H. Yan, A. Zhang, P. Xu, S.H. Pan, M.I. Vai, Y. Gao, Emotion recognition based on multimodal physiological signals using spiking feed-forward neural networks, Biomed. Signal Process. Control 91 (2024).
[55] S. Prabhakaran, N.K. Ayyamperumal, Automatic detection of artifacts and improved classification models for emotional activity detection from multimodal physiological data, J. Intell. Fuzzy Syst. 45 (5) (2023) 8915–8929.
[56] A. Pradhan, S. Srivastava, Hierarchical extreme puzzle learning machine-based emotion recognition using multimodal physiological signals, Biomed. Signal Process. Control 83 (2023).
[57] A. Pradhan, S. Srivastava, Hybrid densenet with long short-term memory model for multi-modal emotion recognition from physiological signals, Multimed. Tools Appl. 83 (12) (2024) 35221–35251.
[58] C. Zou, Z. Deng, B. He, M. Yan, J. Wu, Z. Zhu, Emotion classification with multi-modal physiological signals using multi-attention-based neural network, Cogn. Comput. Syst. 6 (1-3) (2024) 1–11.
[59] M. Yan, Z. Deng, B. He, C. Zou, J. Wu, Z. Zhu, Emotion classification with multichannel physiological signals using hybrid feature and adaptive decision fusion, Biomed. Signal Process. Control 71 (2022).
[60] M.D. Hssayeni, B. Ghoraani, Multi-modal physiological data fusion for affect estimation using deep learning, IEEE Access 9 (2021) 21642–21652.
[61] W. Wu, E. Zuo, W. Zhang, X. Meng, Multi-physiological signal fusion for objective emotion recognition in educational human-computer interaction, Front. Public Health 12 (2024) 1492375.
[62] A. Khaleghi, K. Shahi, M. Saidi, N. Babaee, R. Kaveh, A. Mohammadian, Linear and nonlinear analysis of multimodal physiological data for affective arousal recognition, Cogn. Neurodyn 18 (5) (2024) 2277–2288.
[63] Y. Wu, M. Daoudi, A. Amad, Transformer-based self-supervised multimodal representation learning for wearable emotion recognition, IEEE Trans. Affect. Comput. 15 (1) (2024) 157–172.
[64] D. Pollreisz, N. TaheriNejad, A simple algorithm for emotion recognition, using physiological signals of a smart watch, 2017 39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), (2017), pp. 2353–2356.
[65] W. Kuang, S. Jin, D. Wang, Y. Zheng, Y. Zou, K. Wu, EmoTracer: A User-independent Wearable Emotion Tracer with Multi-source Physiological Sensors Based on Few-shot Learning, in: M. Cannataro, H. Zheng, L. Gao, J. Cheng, J.L. de Miranda, E. Zumpano, X. Hu, Y.-R. Cho, T. Park (Eds.), Proc. - IEEE Int. Conf. Bioinform. Biomed., BIBM, Institute of Electrical and Electronics Engineers Inc., 2024, pp. 2680–2687.
[66] J.A. Domínguez-Jiménez, K.C. Campo-Landines, J.C. Martínez-Santos, E.J. Delahoz, S.H. Contreras-Ortiz, A machine learning model for emotion recognition from physiological signals, Biomed. Signal Process. Control 55 (2020) 101646.
[67] F.P. Akbulut, Hybrid deep convolutional model-based emotion recognition using multiple physiological signals, Comput. Methods Biomech. Biomed. Eng. 25 (15) (2022) 1678–1690.
[68] T. Kim, S. Kim, M. Lee, Y. Kang, S. Hwang, Assessing human emotional experience in pedestrian environments using wearable sensing and machine learning with anomaly detection, Transp. Res. Part F Traffic Psychol. Behav. 109 (2025) 540–555.
[69] A. Arsalan, M. Majid, I.F. Nizami, W. Manzoor, S.M. Anwar, and J. Ryu.Human stress assessment: A comprehensive review of methods using wearable sensors and non-wearable techniques. arXiv preprint arXiv:2202.03033, 2022.
[70] M.B.I. Reaz, M.S. Hussain, F. Mohd-Yasin, Techniques of emg signal analysis: detection, processing, classification and applications, Biol. Proced. Online 8 (1) (2006) 11–35.
[71] A. Rehman, M. Mustafa, I. Israr, M. Yaqoob, Survey of wearable sensors with comparative study of noise reduction ecg filters, Int. J. Com. Net. Tech. 1 (1) (2013) 61–81.
[72] S. Karpagachelvi, M. Arthanari, and M. Sivakumar.Ecg feature extraction techniques-a survey approach. arXiv preprint arXiv:1005.0957, 2010.
[73] A.K. Singh, S. Krishnan, Ecg signal feature extraction trends in methods and applications, Biomed. Eng. OnLine 22 (1) (2023) 22.
[74] P. Koval, P. Kuppens, N.B. Allen, L. Sheeber, Getting stuck in depression: The roles of rumination and emotional inertia, Cogn. Emot. 26 (8) (2012) 1412–1427.
[75] M.K. Moser, M. Ehrhart, B. Resch, An explainable deep learning approach for stress detection in wearable sensor measurements, Sensors 24 (16) (2024) 5085.
[76] L. Paz-Arbaizar, J. Lopez-Castroman, A. Artés-Rodríguez, P.M. Olmos, D. Ramírez, Emotion Forecasting: A Transformer-Based Approach, J. Med. Internet Res. 27 (2025) e63962.
[77] P. Rainville, A. Bechara, N. Naqvi, A.R. Damasio, Basic emotions are associated with distinct patterns of cardiorespiratory activity, Int. J. Psychophysiol. 61 (1) (2006) 5–18.
[78] S. Saganowski, B. Perz, A.G. Polak, P. Kazienko, Emotion Recognition for Everyday Life Using Physiological Signals From Wearables: A Systematic Literature Review, IEEE Trans. Affect. Comput. 14 (3) (2023) 1876–1897.
[79] N. TaheriNejad, D. Pollreisz, Assessment of physiological signals during happiness, sadness, pain or anger, International Conference on Wireless Mobile Communication and Healthcare, Springer, 2016, pp. 107–114.
[80] P. Ekman, An argument for basic emotions, Cogn. Emot. 6 (3-4) (1992) 169–200.
[81] E. Kannegieser, D. Atorf, J. Herold, Measuring flow, immersion and arousal/valence for application in adaptive learning systems, International Conference on Human-Computer Interaction, Springer, 2021, pp. 62–78.
[82] V. Khullar, R.G. Tiwari, A.K. Agarwal, S. Dutta, Physiological signals based anxiety detection using ensemble machine learning, Cyber Intelligence and Information Retrieval: Proceedings of CIIR 2021, Springer, 2022, pp. 597–608.

[83] J.A. Russell, A circumplex model of affect, J. Personal. Soc. Psychol. 39 (6) (1980) 1161.
[84] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, Ł. Kaiser, I. Polosukhin, Attention is all you need, Adv. Neural Inf. Process. Syst. 30 (2017).
[85] A. Chaddad, Y. Wu, R. Kateb, A. Bouridane, Electroencephalography signal processing: A comprehensive review and analysis of methods and techniques, Sensors 23 (14) (2023) 6434.
[86] T. Song, S. Liu, W. Zheng, Y. Zong, Z. Cui, Instance-adaptive graph for eeg emotion recognition, Proceedings of the AAAI Conference on Artificial Intelligence 34 (2020), pp. 2701–2708.
[87] F. Larradet, R. Niewiadomski, G. Barresi, D.G. Caldwell, L.S. Mattos, Toward emotion recognition from physiological signals in the wild: approaching the methodological issues in real-life data collection, Front. Psychol. 11 (2020) 1111.
[88] P. Kuppens, P. Verduyn, Emotion dynamics, Curr. Opin. Psychol. 17 (2017) 22–26.
[89] R.A. Calvo, S. D'Mello, Affect detection: an interdisciplinary review of models, methods, and their applications, IEEE Trans. Affect. Comput. 1 (1) (2010) 18–37.
[90] K. Furukawa, X. Shui, M. Li, D. Zhang, Utilizing tympanic membrane temperature for earphone-based emotion recognition, Sensors 25 (14) (2025) 4411.
[91] C. Fan, K. Zhu, J. Tao, G. Yi, J. Xue, Z. Lv, Multi-level contrastive learning: Hierarchical alleviation of heterogeneity in multimodal sentiment analysis, IEEE Trans. Affect. Comput. 16 (1) (2024) 207–222.
[92] W.-B. Jiang, Z. Li, W.-L. Zheng, B.-L. Lu, Functional emotion transformer for eeg-assisted cross-modal emotion recognition, ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2024, pp. 1841–1845.
[93] F. Chen, J. Shao, S. Zhu, H.T. Shen, Multivariate, multi-frequency and multimodal: Rethinking graph neural networks for emotion recognition in conversation, Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, (2023), pp. 10761–10770.
[94] X. Deng, C. Li, X. Hong, H. Huo, H. Qin, A novel multi-source contrastive learning approach for robust cross-subject emotion recognition in eeg data, Biomed. Signal Process. Control 97 (2024) 106716.
[95] R. Li, Y. Wang, W.-L. Zheng, B.-L. Lu, A multi-view spectral-spatial-temporal masked autoencoder for decoding emotions with self-supervised learning, Proceedings of the 30th ACM International Conference on Multimedia, (2022), pp. 6–14.
[96] A. Sano, R.W. Picard, Stress recognition using wearable sensors and mobile phones, 2013 Humaine association conference on affective computing and intelligent interaction, IEEE, 2013, pp. 671–676.
[97] C. Fan, J. Wang, W. Huang, X. Yang, G. Pei, T. Li, Z. Lv, Light-weight residual convolution-based capsule network for eeg emotion recognition, Adv. Eng. Inform. 61 (2024) 102522.
[98] E. Lazarou, T.P. Exarchos, Predicting stress levels using physiological data: real-time stress prediction models utilizing wearable devices, AIMS Neurosci. 11 (2) (2024) 76.
[99] A. Hongn, F. Bosch, L.E. Prado, J.M. Ferrández, M.P. Bonomini, Wearable physiological signals under acute stress and exercise conditions, Sci. Data 12 (1) (2025) 520.
[100] S. Mu, S. Liao, K. Tao, Y. Shen, Intelligent fatigue detection based on hierarchical multi-scale ecg representations and hrv measures, Biomed. Signal Process. Control 92 (2024) 106127.
[101] Y. Jiang, P. Malliaras, B. Chen, D. Kulić, Real-time forecasting of exercise-induced fatigue from wearable sensors, Comput. Biol. Med. 148 (2022) 105905.